

DL for ENG

LECTURE 9.1

Laminar Flow

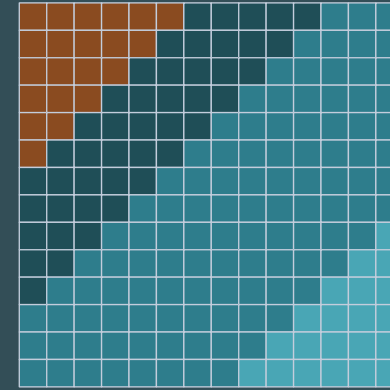
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

MOMENTUM · CONTINUITY · THE FRONT



Re = 20: the front

A flow brings three things at once: a nonlinear term that carries the velocity with itself, a constraint in place of a source, and a pressure with no equation of its own. The Reynolds number says how thin the features get, and that is what decides whether a network can follow them.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Flow equations** — momentum, continuity, the Reynolds number
- 2 **Incompressibility** — a constraint, and three ways to impose it
- 3 **Solving a flow** — the outputs, the points, the checks, the ceiling

Liu carries the method and has no flow chapter, so the physics is the exercise's own: Ex_09.1's coupled Burgers equations, with an exact solution.

READ, AND WHAT EACH ANSWERS

- | | |
|----------------------|------------------------------------|
| Liu ch. 2.2 | which residuals does a flow carry? |
| Liu ch. 2.5.3 | how is a constraint built in? |
| Liu ch. 5.6 | where must the points go? |

Recap

1 – 2 · TIME

1 What does the heat equation add to steady conduction, and how does a PINN treat time?

A storage term, $\rho c_p \partial T / \partial t$, and so one initial field. The diffusivity $\alpha = k / \rho c_p$ says how fast a change spreads and sets the time constant τ . For a PINN time is one more input: the whole block of space and time is trained at once, and the solution can be read at any instant. *Liu ch. 4.1 – 4.3; Ex_08.2 sections 3 – 5*

2 What is causality, and how is it put back into training?

A solution at one instant follows from the earlier ones, but the loss treats all instants alike, so the network can fit late times first and carry an error at the start. First build the initial field into the trial solution; then causal weights, a split window, or more early points. *Liu ch. 4.2 – 4.4; Ex_08.2 section 5*

3 – 4 · PLATE AND INVERSE

3 How is the transient plate set up, and how is its solution checked?

The trial solution $T(x, t)$ holds the initial field and the hole wall at once; the insulated edges stay a loss term. Report the relative and the absolute error, follow the hottest point in time, and use the energy balance once the field is steady. *Liu ch. 4.6; Ex_08.2 sections 4 – 5*

4 How does an inverse problem differ from a forward one?

A parameter of the equation, here the diffusivity α , is trained beside the weights so that the solution fits measured data. It is optimised as $\log \alpha$, so it stays positive. Only the early part of the cooling curve constrains it: once the field has decayed, the data says nothing. *Liu ch. 4.6; Ex_08.2 section 6*

Navier–Stokes

Incompressible flow — two equations for one field: momentum, which says how the velocity changes, and continuity, which says the velocity cannot compress the fluid. The pressure has no equation of its own; it is whatever makes continuity hold.

MOMENTUM, AND A CONSTRAINT

$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = u_x + v_y = 0$$

$\mathbf{u} = (u, v)$ the velocity, p the pressure, ρ the density, ν the kinematic viscosity

THE FLOW RESIDUALS IN PYTORCH

```
u, v, p = split(net(xyt))          # three fields from one network
F_u = grad(u,t) + u*grad(u,x) + v*grad(u,y) + grad(p,x)/rho - nu*lap(u)
F_v = grad(v,t) + u*grad(v,x) + v*grad(v,y) + grad(p,y)/rho - nu*lap(v)
F_c = grad(u,x) + grad(v,y)        # continuity, the constraint
```

WHAT EACH TERM DOES

term	what it does
$\mathbf{u}_t$	unsteady: the flow changes
$(\mathbf{u} \cdot \nabla) \mathbf{u}$	convection: nonlinear, couples u and v
∇p	whatever enforces the constraint
$\nu \nabla^2 \mathbf{u}$	viscosity: smooths, and sets the scales

The second row is the one that is new. Everything before this block was linear in the field; a velocity that carries itself is not.

The Reynolds Number

Reynolds number — the ratio of convection to viscosity, $Re = UL/\nu$. Scaled by the device size and the inlet speed, it is the only parameter left in the equations, and it sets how thin the features of the solution are: in Ex_09.1 the front's width is $8/Re$.

ONE NUMBER, AND IT SETS A LENGTH

$$Re = \frac{UL}{\nu} \quad \delta = 8\nu = \frac{8}{Re}$$

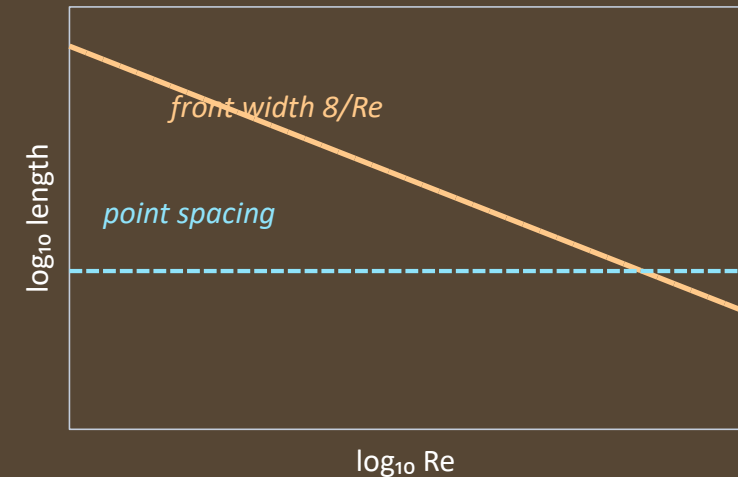
$$\mathbf{u}_t^* + (\mathbf{u}^* \cdot \nabla) \mathbf{u}^* = -\nabla p^* + \frac{1}{Re} \nabla^2 \mathbf{u}^*$$

U the inlet speed, L the device size, δ the width of the front; starred: scaled by U and L

THE CEILING, BEFORE TRAINING

```
nu = 1.0 / Re          # scaled: U = L = 1, so nu is 1/Re
delta = 8 * nu          # Ex_09.1's front width
print(delta / spacing(n_points)) # points across the front: want > 5
```

FRONT WIDTH AGAINST Re



Cyan is the spacing of 4 000 points on a unit square. Where the amber line crosses it, the front is thinner than the sample: the ceiling of Ex_09.1.

Coupled Burgers

Coupled Burgers equations — Navier–Stokes with the pressure and the constraint removed: two equations, still nonlinear, still coupled through the convective term. They keep everything that makes a flow hard except the constraint that makes it slow, and they have an exact solution.

THE STEPPING STONE OF EX_09.1

$$u_t + u u_x + v u_y = \nu (u_{xx} + u_{yy})$$

$$v_t + u v_x + v v_y = \nu (v_{xx} + v_{yy})$$

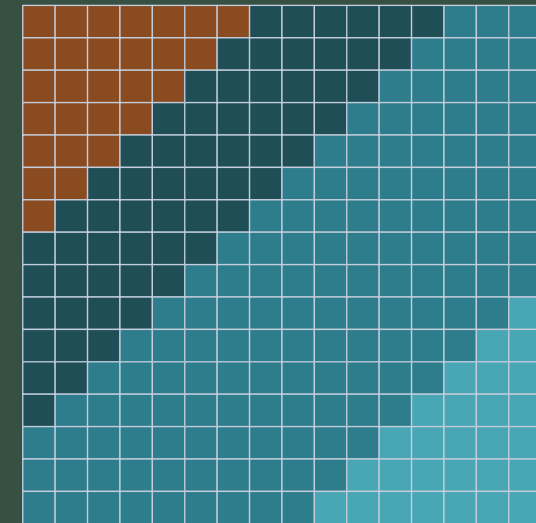
$$u = \frac{3}{4} - \frac{1}{4} \sigma\left(\frac{y - x - t/4}{8\nu}\right), \quad u + v = \frac{3}{2}$$

σ the logistic function, ν the viscosity; the front lies along $y = x + t/4$

COUPLED BURGERS IN PYTORCH

```
u, v = split(net(xyt))                # two outputs, one network
F_u = grad(u,t) + u*grad(u,x) + v*grad(u,y) - nu*lap(u)  # v enters here
F_v = grad(v,t) + u*grad(v,x) + v*grad(v,y) - nu*lap(v)  # and u here
```

u OVER THE SQUARE, AT t = 0



$Re = 20$: the front is 0.4 wide

the front

along $y = x + t/4$, width $8/Re$

the identity

$u + v = 3/2$, everywhere

Flow Boundary Conditions

Flow boundary conditions — a wall carries no-slip, $\mathbf{u} = 0$, because a viscous fluid does not slide along it. An inlet prescribes the velocity profile, an outlet prescribes the pressure and lets the velocity leave, and the pressure needs one reference value somewhere.

FOUR KINDS OF EDGE

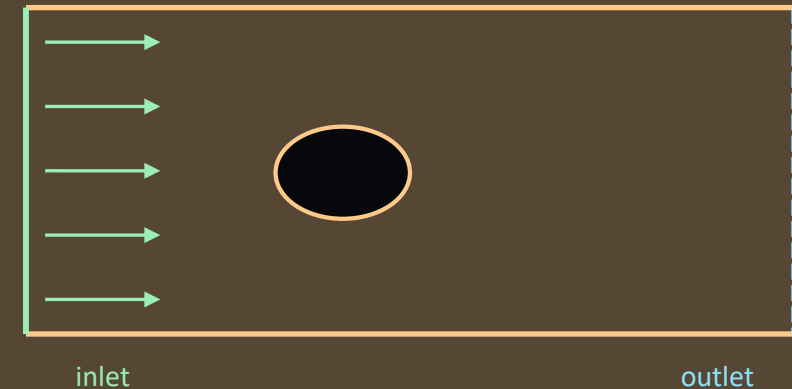
$\mathbf{u} = 0$ on the wall, $u = U(y)$ at the inlet
 $\mathbf{u}$ the velocity, $U(y)$ the inlet profile

THE EDGES IN PYTORCH

```
B_wall = torch.cat([u(x_w), v(x_w)], 1)      # no-slip: both components
B_in = torch.cat([u(x_i) - U_prof, v(x_i)], 1) # the inlet profile
B_out = p(x_o)                                # the outlet reference
L_bc = sum(b.pow(2).mean() for b in (B_wall, B_in, B_out)) # one term each
```

THE CHANNEL OF EX_09.2

walls and obstacle: no-slip



no-slip

both components zero on a wall

inlet

a profile, not a constant

outlet

pressure fixed, velocity free

Imposing Incompressibility

Incompressibility — the constraint $\nabla \cdot \mathbf{u} = 0$, which every instant of the flow must satisfy. It can be asked for as a loss term, built into the network through a stream function, or projected onto after each step. Only the second holds it exactly and for free.

THREE WAYS, AND WHAT EACH COSTS

- **a penalty** — one more term, one more weight
- **a stream function** — exact, and one output fewer
- **a projection** — exact, and a solve at every step

TALK · 2 MIN How would you explain what incompressibility asks of a velocity field, in your own words? Then, in the table on the right, which of the three would you choose for the channel of Ex_09.2, and what does it cost you?

THE THREE ROUTES

route	holds	costs
penalty term	roughly	a weight to tune
stream function	exactly	a derivative more
projection	exactly	a linear solve

The continuity residual has no time derivative and is the smallest term in the sum, so as a penalty it is the term most easily outvoted. That is the argument for building it in.

Stream Function

Stream function — a single scalar ψ whose derivatives are the velocity: $u = \psi_y$ and $v = -\psi_x$. Continuity then holds identically, because the two cross derivatives cancel, so the network outputs ψ instead of u and v and the loss has no continuity term at all.

ONE OUTPUT INSTEAD OF TWO

$$u = \psi_y, \quad v = -\psi_x$$

$$\nabla \cdot \mathbf{u} = \psi_{yx} - \psi_{xy} \equiv 0$$

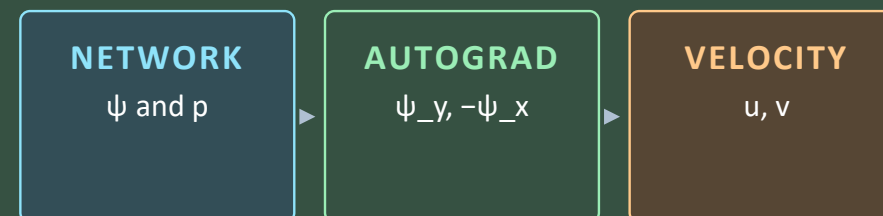
ψ the stream function, u, v the velocity components it gives

- **the price** — one derivative more in every residual
- **the gain** — no weight, and no residual to balance

THE STREAM FUNCTION IN PYTORCH

```
psi, p = split(net(xy))          # one scalar, plus the pressure
u, v = grad(psi, y), -grad(psi, x) # continuity holds identically
```

FROM ONE SCALAR TO A FLOW



outputs	ψ and p , not u, v and p
continuity	gone from the loss
derivatives	third order in the residual

The momentum residual differentiates u twice and u is already a derivative of ψ , so the graph is one order deeper. That is the whole price.

Pressure Reference

Pressure reference — only the gradient of the pressure appears in the momentum equation, so adding a constant to p leaves the flow unchanged. A network will drift along that freedom unless one value is fixed: a reference point, or the pressure at the outlet.

THE SAME FLOW, A DIFFERENT LEVEL

$p \rightarrow p + c \Rightarrow$ the same velocity field

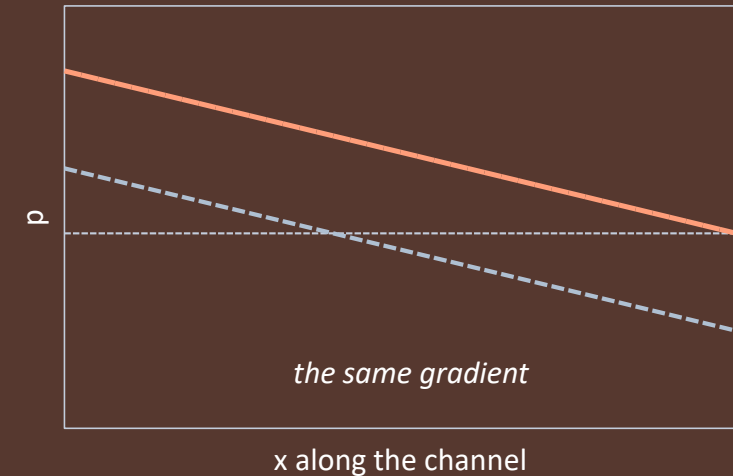
p the pressure, c any constant: only ∇p enters the equations

- **the symptom** — a pressure that wanders during training
- **the fix** — one value pinned, usually at the outlet

THE REFERENCE IN PYTORCH

```
L_ref = p(x_out).pow(2).mean()      # the outlet is the reference
loss = L_mom + L_bc + w_ref * L_ref  # one term fixes the level
print(p(xy).mean().item())          # watch it: it should not drift
```

TWO PRESSURES, ONE FLOW



Both drive exactly the same flow. Only the gradient is physical, so quote a pressure difference, never a pressure.

Shared Network

Shared network — one network with several outputs, rather than one network per field. The hidden layers are shared, so the coupling between the components is represented inside the network, and the parameter count is spent once instead of twice.

SHARED LAYERS, SEVERAL HEADS

$$(\psi, p) = \mathcal{N}_{\hat{w}}(x, y, t)$$

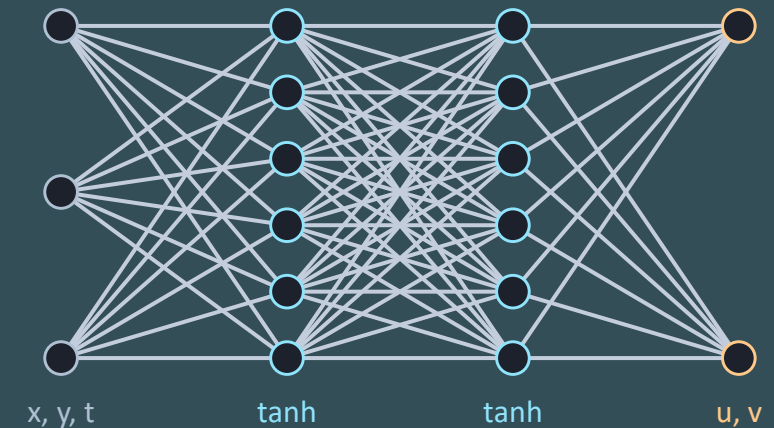
ψ the stream function, p the pressure, $\mathcal{N}_{\hat{w}}$ one network for both

- **shared** — the coupling is learned once
- **separate** — the same budget spent twice

THE NETWORK IN PYTORCH

```
net = MLP(n_in=3, n_hidden=64, n_layers=6, n_out=2) # (x, y, t) in, two out
u, v = net(xyt).split(1, dim=1) # the two components
loss = F_u.pow(2).mean() + F_v.pow(2).mean() + L_bc # one loss, two residuals
```

ONE TRUNK, TWO OUTPUTS



Notebook 02 compares this against two separate networks at the same parameter count, and the coupled term is the reason the shared one wins.

Where the Points Go

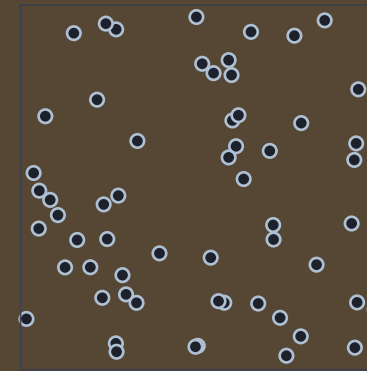
Sampling a flow — a flow puts its difficulty in thin places: a front, a boundary layer, a wake. A uniform sample spends almost every point where the solution is flat, so points are graded towards the walls and refined where the residual is largest.

THE POINTS BELONG WHERE THE FIELD BENDS

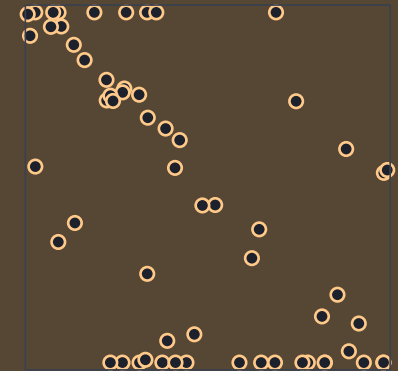
- **uniform** — the error plateaus, whatever you add
- **near the wall** — graded, because the layer is thin
- **at the front** — refined where the residual is large

TALK · 2 MIN How would you decide where a solver should spend its effort, in your own words? Then, in the two squares on the right, which sample would you trust at $Re = 500$, and what would you measure to be sure?

THE SAME BUDGET, TWO WAYS



uniform



refined at the front

Sixty points each. On the left almost all of them sit where u is flat and already right; on the right they follow the diagonal, where it is not.

Checking a Flow

Checking a flow — a flow gives several checks that need no exact solution: the continuity residual, an identity the solution must obey, the mass through any cross-section, and a profile against a reference. A low loss with a smooth wrong field passes none of them.

FOUR CHECKS, NONE OF THEM THE LOSS

$$\mathcal{L} = \mathcal{L}_u + \mathcal{L}_v + w_c \mathcal{L}_{cont} + \mathcal{L}_{BC} + \mathcal{L}_{IC}$$

$\mathcal{L}_u, \mathcal{L}_v$ the momentum residuals, $\mathcal{L}_{cont}$ continuity, w_c its weight

- **the loss** — says the residual is small, and nothing else
- **the checks** — say the field is a flow

THE CHECKS IN PYTORCH

```
div = (grad(u,x) + grad(v,y)).abs().mean()    # continuity, after training
ident = (u + v - 1.5).abs().max()              # Ex_09.1's identity
mass = trapz(u(x0, ys), ys)                   # mass through a section
err = (pred - exact).norm() / exact.norm()     # only where there is an exact
```

WHAT EACH CHECK CATCHES

check	catches
continuity	a field that is not a flow
the identity	a front in the wrong place
mass balance	a leak through a wall
a profile	the wrong shape, at the right size

A smooth, plausible, wrong field is more dangerous than a crash, because nothing in the training log looks unusual.

The Reynolds Ceiling

Reynolds ceiling — the Reynolds number at which a given sample stops resolving the solution. Below it the error falls with training; above it the front is thinner than the spacing between points, the loss still falls, and the field is smooth and wrong.

THE LOSS KEEPS FALLING, THE FIELD DOES NOT

- **below the ceiling** — more points, less error
- **above it** — more points, the same error
- **the fix** — refine at the front, or drop Re

THE SWEEP IN PYTORCH

```
for Re in [20, 50, 100, 200, 500]:  
    net = train(nu=1/Re, n_points=4000)  # notebook 03's sweep  
    log(Re, 8/Re, rel_error(net))        # everything else fixed  
                                         # width, and what it cost
```

SCHEMATIC · ERROR AGAINST Re



Schematic: the knee sits where the front width $8/Re$ falls below the point spacing. Notebook 03 measures where yours is.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
adaptive refinement	points where the residual is large	Liu ch. 5.6	thin fronts, high Re
time windows	the transient in pieces	Liu ch. 4.2 – 4.3	long unsteady runs
inverse flow	viscosity or pressure from data	Liu ch. 2.2	measured velocity fields
<u>porous media</u>	Darcy flow: no convection	Liu ch. 2.4	groundwater, CO ₂ storage
turbulent mean flow	an effective viscosity	Pope ch. 4, 10	any real Reynolds number

WHY THEY ARE LEFT OUT

Ex_09.1 is steady work at a Reynolds number you choose, with an exact solution to score against, so none of them is needed to do it. The last row is the next lecture, and it is the row that decides whether any of this reaches a real flow.

Exercise 9.1

THE NOTEBOOKS, IN ORDER

- **00** — the problem, its exact solution, its front
- **01** — the two residuals, written by you, and trained
- **02** — the control panel: points, width, depth, L-BFGS
- **03** — the Reynolds sweep, up to the ceiling
- **04** — compares them, and the report

THE QUESTION EACH NOTEBOOK ANSWERS

- 01** where is the error largest?
- 02** what do more points buy?
- 03** where is your ceiling?

REPORT QUESTION, TO CONCLUDE

Your sweep found a Reynolds number where the model stopped following the flow. Argue from the front width and your point spacing why it sits there, and say what you would change first to move it.

Key Takeaways

NAVIER-STOKES

what it is	momentum, plus a constraint
function	the velocity, and a pressure that enforces
train, tune	scale first; Re is the only number left
vs heat	nonlinear, coupled, constrained
used for	channels, wakes, anything that moves

SOLVING A FLOW

what it is	one network, several outputs
function	ψ for continuity, p for the level
train, tune	points at the front and the wall
vs uniform points	the error stops falling
used for	up to the ceiling $8/Re$ sets

Next — L9.2: the same equations at a Reynolds number no sample can resolve.

Questions

1 – 2 · NAVIER–STOKES

1 What are the incompressible flow equations, and what does the pressure do?

Momentum, which says how the velocity changes, and continuity, which says the fluid is not compressed. The convective term is nonlinear and couples the components. The pressure has no equation of its own: it is whatever makes continuity hold, and is known only up to a constant. *Liu ch. 2.2; Ex_09.1 notebooks 01 – 02*

2 What does the Reynolds number tell you, and how is incompressibility imposed?

It is the ratio of convection to viscosity, the only parameter left after scaling, and it sets how thin the features of the flow are. Continuity can be a loss term, which is easily outvoted, or be built in by a stream function, which is exact at the price of one more derivative. *Liu ch. 2.5.3; Ex_09.1 notebook 02; Ex_09.2 notebook 01*

3 – 4 · SOLVING A FLOW

3 How is a flow network built, and where do its points go?

One network with several outputs, so that the coupling is represented in the shared layers. A flow keeps its difficulty in thin places, a front or a boundary layer, so the points are graded towards them; uniform points stop helping once the front is thinner than their spacing. *Liu ch. 2.1, 5.6; Ex_09.1 notebooks 02 – 03*

4 How do you check a flow that has no exact solution?

By the continuity residual, the mass through any cross-section and a profile against a reference. Above the Reynolds number its points can resolve, the loss still falls and the field is smooth and wrong. That is worse than a crash, because nothing warns you. *Ex_09.1 notebooks 01, 03*

NEXT

L9.2 — Turbulent Flow

What happens above the ceiling: averaged equations, a closure that has to be modelled, and the one use where the method is clearly the right tool.

BEFORE THE NEXT SESSION

read	Liu ch. 2.5.3, 5.6
run	Ex_09.1 notebooks 00 to 03
bring	your ceiling, and the front width there
think about	the thinnest feature in your problem

*A laminar flow is solved, and its limit is measured.
Next, the flows that are above that limit by
construction.*